

Literature Gap Audit and Next-Step Plan

SOH Estimation of Lithium-Ion Batteries on the NASA Dataset — Why Our Scores Differ from Published Results, a 2022–2026 Literature Comparison, and Recommended Next Steps

Prepared for: Don (M11452803), Master's student | **Advisor:** Y.F. Luo, NTUST | **Date:** 26 September 2026
Basis: Round 2 Progress Report (Steps 1–18, 25 Sep 2026) and a literature search of 2022–2026 journal papers using the NASA battery dataset.

Summary in one paragraph. Our lower scores are **mainly caused by a stricter evaluation protocol and different input features, not by a data-processing error**. Almost every NASA paper that reports $R^2 > 0.95$ uses at least one of: (a) a random or chronological split *inside the same battery*; (b) inputs that are indirect capacity proxies (previous-cycle capacity, cycle index, accumulated Ah, discharge duration); or (c) 34 NASA cells with R^2 pooled over many batteries. Under a strict unseen-battery protocol with only B0005/06/07/18, recent 2026 papers report $R^2 \approx 0.53\text{--}0.88$ and $\text{RMSE} \approx 4\text{--}7\%$ SOH. **Our TD-BPNN result ($R^2 \approx 0.85$, RMSE 3.59 %SOH) is at the same level or slightly better.** However, $R^2 = 0.85$ is not yet a novelty claim: (1) the time-domain features have not been audited for capacity proxies; (2) the final model may have been trained on only two batteries; and (3) a 2026 paper has already combined EIS + discharge features + cross-battery evaluation (Section 3).

Important — this material can go directly into your thesis. The comparison table, the protocol analysis, and the reference list in this document are written so that you can reuse them in your thesis (Literature Review, Methodology, Results and Discussion chapters). Section 6 explains exactly where each part belongs. Before you cite any number, please open the full text of the paper and confirm the value, the unit, and the split protocol yourself.

1. Current Status of Your Project (Round 2)

Item	Current result
Data	NASA BatteryAgingARC-FY08Q4: B0005, B0006, B0007, B0018 (168/168/168/132 discharge cycles)
Protocol	Nested 4-fold leave-one-battery-out (LOBO); inner validation = first remaining battery; 3 seeds (42, 7, 123); fold-train-only standardization; leakage probes all PASS
SOH definition	$\text{SOH} = Q / Q_{\text{ref}}$, Q_{ref} = mean of the first five discharge capacities (in %SOH)
Best model	Time-domain (TD) features → BPNN: RMSE 3.59 %SOH, $R^2 \approx 0.85$, positive R^2 on all four folds (B0018 ≈ 0.79)
Other pipelines	TD-Ridge 4.35; TD + EIS fusion 4.89 (R^2 0.726); EIS-only pipelines 8.3–8.6; mean predictor 11.5 (RMSE, %SOH)
Negative results	AE on EIS does not beat PCA; AE on TD features gives R^2 0.43 vs 0.85 for raw TD features
Declared limitations	Per-point EIS frequency not recoverable (0.1 Hz–5 kHz grid is an assumption); TD features use a completed discharge cycle (cycle-level SOH, not mid-cycle); no ICA/DVA/pulse-resistance features

2. Question 1 — Why Are Our Scores Lower on the Same NASA Data?

We checked the capacity data first. The four batteries have 168/168/168/132 cycles and minimum SOH values of 69.9 / 57.2 / 74.4 / 72.9 %, which agree with the original NASA files. We did not find a data-processing error. The differences come from the factors below, ordered by expected impact.

#	Factor	Our study	Typical high-score papers	Effect on scores
1	Train/test split	Nested LOBO: the test battery is never seen during training or tuning	First 50–80 % of cycles of the <i>same</i> battery for training, the rest for testing; or a random split [1],[3],[5],[6],[7]	Largest. In the same paper, changing a random split to a battery-wise split reduced R^2 from 0.979 to 0.786 [11]
2	Capacity-proxy inputs (label shortcut)	TD feature list not yet audited	Previous-cycle capacity $Q(k-1)$ [3]; previous-cycle maximum capacity [4]; cycle number, discharge duration [11]; accumulated charge, cycle index [10]	Very large. Under constant-current discharge, $\text{Capacity} = I \times t$, so the model is effectively given the answer
3	Number of training batteries	Only 4 batteries; inner validation = one fixed battery. If the model is not re-fitted after tuning, the final model is trained on only 2 batteries	34 NASA cells, 24 for training [13]; 34 cells, 1,830 samples [11]	Large. Two batteries cannot represent cell-to-cell variation
4	Extrapolation (B0006)	B0006 falls to 57 %SOH; the other three stop at about 70–74 %	Not an issue with same-battery splits	The B0006 fold requires extrapolation. Chen et al. [9] also report only $R^2 = 0.25 \rightarrow 0.53$ for B0006
5	How R^2 is computed	Per battery (narrow SOH range, std 8–12 %SOH), then averaged	One R^2 pooled over many batteries	Pooling adds between-battery variance to the denominator, so R^2 looks higher for the same RMSE
6	SOH definition and units	Q/Q_{ref} (first 5 cycles), in %SOH	$Q/2.0 \text{ Ah}$ (nominal), Q/Q_0 , or errors in Ah or as a 0–1 fraction	An RMSE of 0.02 looks small but equals only about 1–2 %SOH. Units must be unified before comparing
7	Different discharge cut-off voltages	B0005 2.7 V, B0006 2.5 V, B0007 2.2 V, B0018 2.5 V	No effect with same-battery splits	Features such as minimum voltage, voltage drop, and full discharge time have different scales per battery $\rightarrow$ domain shift across batteries

Points Don should check in the repository

- Does the TD model use **all 636 discharge cycles**, or only the cycles aligned with EIS (tolerance 1 keeps only about 51 %)?
- TD-Ridge reaches RMSE 4.35. If full discharge duration were an input, a linear model should be almost perfect, so a direct duration feature is *probably* not present — but every feature must still be listed and checked (Section 5-B).
- After inner validation selects the hyper-parameters, is the model re-trained on **all three** training batteries?

3. The Closest Published Paper (Verified)

S. A. Sardar, F. Akhtar, W. Ali, W. Y. Kim, “Multi-Source Impedance and Discharge Feature Learning for Cross-Battery State-of-Health Estimation of Lithium-Ion Batteries,” *Batteries*, vol. 12, no. 8, p. 291, 2026. DOI: 10.3390/batteries12080291 (published 6 Aug 2026).

Aspect	Sardar et al. 2026 [11]	This thesis (current)
Data	NASA, 34 cells, 1,830 EIS-SOH pairs	NASA, 4 cells (B0005/06/07/18)
EIS features	Re/Im statistics, slopes, Nyquist area, resampled points, NASA-provided Re and Rct	Rectified_Impedance $\rightarrow$ PCHIP 256-dim $\rightarrow$ PCA / AE
Time-domain features	Mean voltage, voltage drop, discharge duration , current/temperature, cycle number , EIS index	TD features (audit pending)
Model	RF / ET / GB / HGB / XGBoost	BPNN, Ridge
Validation	Random split; battery-wise (7 unseen cells); repeated battery-wise (5 partitions)	Nested 4-fold LOBO $\times$ 3 seeds with leakage probes
Results	Random: R^2 0.979. Strict battery-wise: R^2 0.786, RMSE 7.37 %. Repeated: R^2 0.881	R^2 0.85, RMSE 3.59 %SOH

Aspect	Sardar et al. 2026 [11]	This thesis (current)
Stated limitations	Discharge duration and cycle number act as indirect SOH proxies; EIS resampling not frequency-aligned	EIS frequency is an assumption; cycle-level only

Assessment. The combination “EIS + discharge features + cross-battery evaluation” has already been published, so it cannot be our novelty. At present, our differences are a stricter nested protocol, the PCHIP / frequency-assumption analysis, and complete evidence that EIS-only models fail on unseen batteries. These show good research quality, but they are not yet a new contribution. **Sardar et al. admit proxy leakage themselves and do not report a proxy-free version. This is the gap we can target.**

4. Question 2 — Comparison of NASA-Based SOH Studies (2022-2026)

Yellow rows use a strict unseen-battery (LOBO/LOCO) protocol; the green row is this thesis. Bracketed numbers refer to the reference list in Section 7. “–” means the value was not reported in the part of the paper we could access.

#	Study (journal, year)	Data	Algorithm	Input features	Validation	R ²	RMSE	MAE	Notes
1	Zhang D. et al., TCN (Front. Energy Res., 2022)	NASA B5/7/18	TCN	Temperature-variation-rate HI	Same battery, chronological (start at cycle 90)	–	≤1.80 %	≤1.46 %	Same-battery split
2	Dong H. et al., improved GPR (Int. J. Electrochem. Sci., 2022)	Two public datasets	GPR + combined kernel + PCA	dQ/dV, dV/dT features	Not stated	–	error <1.5 %	–	Abstract-level information only
3	Xu G. et al., STL-LSTM (PLOS ONE, 2024)	NASA B5/6/7	LSTM + spatio-temporal attention	Charge time in 3.9–4.0 / 4.0–4.1 V; moving averages of capacity; Q(k–1)	Same battery, first 50/70 %	–	~0.016–0.019 Ah	~0.012–0.015 Ah	Uses previous-cycle capacity
4	Giuliano A. et al., Transformer TL (Energies, 2025)	NASA 34 cells → Oxford	Transformer transfer learning	V/I/T/time, cycle number , previous-cycle capacity	Pre-train on NASA; 1-epoch fine-tune on Oxford	–	0.0146	–	Cross-dataset, but with proxies
5	Salem N. et al., short-term discharge (Energy Reports, 2026)	NASA 4 cells	LSTM / BiLSTM-MHSA / CNN / CatBoost	Per-cycle mean V/I/T; 10/20/30-min discharge windows	Chronological 80/20	0.90 (full); 0.94 (10 min)	0.030 / 0.023	0.014 / 0.011	Partial window, no cross-battery test
6	Okour M. et al., limited voltage (JLPEA, 2026)	NASA 4 cells + PyBaMM	RF / FNN	N early-discharge voltage samples (1-min sampling)	Random 80/20	0.92–0.96	1.28–2.87 %SOH	–	Only 5 voltage points, but random split
7	Zhao Y. et al., GAF-CNN-LSTM (Sci. Rep., 2026)	NASA 4 cells + Oxford	GAF-CNN + cross-attention + LSTM	IC curves from CC charging	Same battery, 50 % training	~0.995	≤0.004	≤0.004	Same-battery split
8	Mohamud N. M. et al., BiLSTM-RF (Batteries, 2026)	NASA 4 cells + Oxford	Attention-BiLSTM + RF	Wavelet-denoised V/I/T, ICA, PCA	LOBO-CV	–	0.0229 (B0007)	–	LOBO; only partial results listed
9	Chen B. et al., physics-consistent domain-aware (Batteries, 2026)	NASA B5/6/7	Monotone LightGBM + covariance alignment + conformal	13 HIs (CC-charge time, IC-peak voltage, CV time...) + cycle index	LOCO (2 train, 1 test)	B0006: 0.25→0.53	mean 0.0499→0.0425	–	Closest to our conditions; similar level
10	Zhao J. et al., EWDC (Energies, 2026)	NASA 4 + CALCE CS2 4	GCN + recursive degradation increments + consistency constraints	17 partial CC-CV charge HIs (incl. accumulated charge , cycle index)	LOBO (within each dataset)	0.975	1.21 %	0.65 %	Recursion from $\hat{y}_1=1$ gives a monotonic prior; averaged over 8 cells
11	Sardar S. A. et al. (Batteries, 2026)	NASA 34 cells	RF / ET / GB / XGB	EIS statistics + discharge duration + cycle	Random / battery-wise	0.979 / 0.786–0.881	2.18 % / 7.37 %	3.22 % (RF, repeated)	Closest to this thesis
12	Meng Y. et al., HSSA-Mamba (Batteries, 2026)	NASA (11/5/4-cell scales)	Mamba + Q-former fusion	Discharge V/I/T sequence + EIS	Not detailed; out-of-sample B28	–	–	0.89 / 1.46 / 2.71 %	MAE 2.7 % at the 4-cell scale

#	Study (journal, year)	Data	Algorithm	Input features	Validation	R ²	RMSE	MAE	Notes
13	Shi X. et al., DS-Transformer (Sci. Rep., 2026)	NASA 34 cells, 2,794 pairs	Dual-stream Transformer + cross-attention	Full-discharge V/I/T sequence (~490 steps) + Re/Rct	Battery-wise 24/5/5 cells	0.978 (pooled)	1.67 %	1.24 %	Full sequence implicitly contains discharge time; pooled R ²
—	This thesis: TD-BPNN	NASA 4 cells	BPNN	TD features (audit pending)	Nested 4-fold LOBO × 3 seeds	0.85	3.59 %SOH	–	Strictest protocol

MRE/MAPE: most papers do not report it (DS-Transformer: MAPE 1.51 %), so it is not a separate column. Please add MAPE to your own results so that readers can compare. **Journal quartiles (Q1/Q2)** are not listed; check them in JCR/SJR. A Sci. Rep. 2025 paper (“Cycle based state of health estimation of lithium ion cells using deep learning architectures”) could not be accessed (rate-limited) and is not included yet.

Pattern in the table	Typical R ²
Same-battery split or random split	0.95–0.99
Strict cross-cell, only 3–4 cells, no capacity-proxy inputs	0.53–0.88
LOBO but still very high (EWDC 0.975; DS-Transformer 0.978)	These use accumulated charge, cycle index, full-discharge sequences, or many pooled cells
This thesis (strictest protocol)	0.85 (RMSE 3.59 %SOH)

5. Question 3 — Recommended Next Steps for Don

A. Fix the evaluation protocol first (within 1 week; may improve scores without changing the model)

- **Re-fit:** after the inner validation selects the hyper-parameters, re-train on **all three** training batteries before testing. Alternatively, make the inner loop a 3-fold LOBO. This should help the B0006 fold most.
- **Report R^2 in two ways:** per-fold mean and pooled. Report RMSE in both %SOH and Ah, and add MAPE. This allows a fair comparison with the literature.
- **Protocol-gap experiment:** with the same features and the same model, run (i) a random split, (ii) a chronological split within each battery, and (iii) nested LOBO. We expect (i) to reach $R^2 > 0.97$. This directly shows that the gap is caused by the protocol, and it can become a full section of the thesis.

B. Target-proxy audit (the most important scientific question)

- List every TD feature and label it *safe* or *proxy*. Proxies include discharge duration, end-of-discharge time, $\int I dt$ (accumulated Ah), cycle index, and previous-cycle capacity.
- Run three settings under the same nested LOBO $\times$ 3 seeds: **TD-All**, **TD-Proxy-Free**, and an **Oracle-proxy baseline** ($I \times t$ only). The oracle should give $R^2 \approx 1$ and explains where high published scores come from.

C. Improve proxy-free accuracy (ordered by expected benefit / effort)

- **Use charging-segment features.** All four batteries use the same charge protocol (CC 1.5 A $\rightarrow$ 4.2 V $\rightarrow$ CV), while their discharge cut-off voltages differ. Use CC-charge time in fixed voltage windows (3.9–4.0 V, 4.0–4.1 V), CV-phase time, and IC-peak voltage/height. These features are naturally aligned across batteries and are partial-window by design (the approach used in [9] and [10]).
- **Fixed voltage window for discharge features**, e.g. 4.0 $\rightarrow$ 3.6 V (all batteries pass through it), instead of whole-cycle minimum voltage or total time.
- **Self-referenced normalization:** divide (or subtract) each feature by the value from the same battery's first cycle. No label is needed at deployment, and it removes initial cell-to-cell offsets (e.g. B0006 starts at 2.035 Ah; the others at about 1.86 Ah). Low cost, potentially large gain.
- **Monotonic / physical prior:** predict the degradation increment ΔSOH and accumulate it (as in EWDC [10]), or add a monotonicity penalty to the BPNN.
- **Window-length curve:** plot RMSE versus window length (5/10/20/30 min, or $\Delta V = 0.1/0.2/0.3$ V). This answers “how little data is enough?” and is the key figure for a partial-cycle contribution.

D. Wider validation

- **Within NASA:** add B0025–B0056 cells with similar conditions so that training grows from 3 to 10+ batteries, and show a learning curve of LOBO performance versus the number of training batteries.
- **Cross-dataset:** develop and freeze the model on NASA, then test directly on CALCE CS2 (the dataset used by EWDC, so results are comparable), and later Oxford if time allows.

E. Role of EIS and the autoencoder

- Replace the 256-dim interpolated spectrum with a few scalars (NASA-provided Re and Rct , $|Z|$ range, arc height) and use them as auxiliary fusion features. Keep fusion in the main method only if it clearly beats TD-only.
- Move the AE to the ablation / negative-result section (AE-TD R^2 0.43 vs TD 0.85). Do not keep the AE just to match the old thesis title.

Suggested new thesis title

Proxy-Controlled, Cross-Battery SOH Estimation from Partial Charge/Discharge Windows: Quantifying the Protocol Gap on the NASA Dataset

(If cross-dataset validation succeeds, “Cross-Battery” can become “Cross-Dataset”.)

Three claimable contributions: (1) quantify how much of the high published accuracy comes from the evaluation protocol and capacity-proxy features (protocol gap + oracle proxy); (2) keep unseen-battery accuracy under proxy-free, partial-window conditions; (3) cross-dataset validation.

Information Don should send back

- The complete definition of every TD feature (code path and formula).
- Whether the model is re-fitted on three batteries after inner selection.
- Per-fold R^2 / RMSE, and the pooled R^2 .
- The number of TD samples, and whether only EIS-aligned cycles are used.

6. How to Use This Material in Your Thesis

All comparisons and references in this document are meant to be **reused in your thesis**. The table below shows where each part fits.

Thesis chapter	What to include from this document
Ch. 1 Introduction	The research gap in Section 3: EIS + discharge + cross-battery already exists [11]; the open problem is proxy-free, partial-window, unseen-battery SOH estimation. State your contributions (Section 5, title box).
Ch. 2 Literature Review	The comparison table in Section 4 (update and extend it to 20–30 papers). Group the studies by (A) time-domain, (B) EIS, (C) fusion, (D) cross-battery, (E) cross-dataset, (F) partial-cycle. Use the “pattern” table to explain why published R^2 values vary so much.
Ch. 3 Methodology	Justify nested LOBO, fold-train-only normalization, leakage probes, the SOH definition, and the proxy-feature definitions (Section 5-B), citing the papers that use weaker protocols as motivation.
Ch. 4 Results	Protocol-gap experiment (random vs chronological vs LOBO), TD-All vs TD-Proxy-Free vs oracle, window-length curve, and a final table placing your results next to the literature (with the same units).
Ch. 5 Discussion & Limitations	Section 2 (why scores differ: split, proxies, training cells, B0006 extrapolation, R^2 pooling, units, cut-off voltages); the EIS frequency assumption; cycle-level vs real-time estimation.
Appendix	Negative results: AE on EIS and on TD features; interpolation ablation; tolerance ablation.

Before citing, please check: (1) open the full paper and confirm every number, unit, and split protocol; (2) cite the original paper, not this summary; (3) when you compare numbers, convert all of them to the same unit (%SOH) and note the protocol next to each value; (4) record the journal quartile from JCR/SJR.

7. References (IEEE style, ready for the thesis)

- [1] D. Zhang, W. Zhao, L. Wang, X. Chang, X. Li, and P. Wu, “Evaluation of the state of health of lithium-ion battery based on the temporal convolution network,” *Frontiers in Energy Research*, 2022, doi: 10.3389/fenrg.2022.929235.
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- [3] G. Xu, J. Xu, and Y. Zhu, “LSTM-based estimation of lithium-ion battery SOH using data characteristics and spatio-temporal attention,” *PLOS ONE*, 2024, doi: 10.1371/journal.pone.0312856.
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- [10] J. Zhao, X. Qian, Y. Zhang, Y. Mu, B. Wang, and H. Xiao, "Unseen-cell SOH prediction via energy-aware warm-up and degradation-consistency constraints," *Energies*, vol. 19, no. 18, p. 4326, 2026, doi: 10.3390/en19184326.
- [11] S. A. Sardar, F. Akhtar, W. Ali, and W. Y. Kim, "Multi-source impedance and discharge feature learning for cross-battery state-of-health estimation of lithium-ion batteries," *Batteries*, vol. 12, no. 8, p. 291, 2026, doi: 10.3390/batteries12080291.
- [12] Y. Meng *et al.*, "Multimodal state of health prediction for lithium-ion batteries via Mamba-based fusion of discharge curves and impedance spectra," *Batteries*, vol. 12, no. 6, p. 196, 2026, doi: 10.3390/batteries12060196.
- [13] X. Shi, J. Zeng, and J. Zhang, "DS-transformer: A dual-stream transformer for lithium-ion battery state-of-health estimation via cross-attention fusion of discharge curves and impedance features," *Scientific Reports*, 2026, doi: 10.1038/s41598-026-52202-6.
- [14] B. Saha and K. Goebel, "Battery data set," NASA Ames Prognostics Data Repository, NASA Ames Research Center, Moffett Field, CA, 2007. [Dataset]